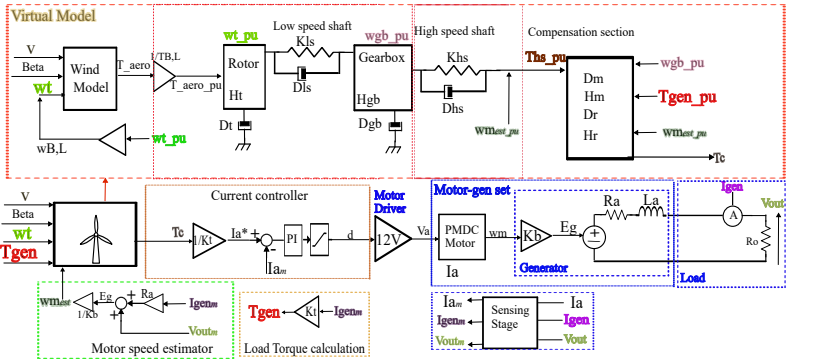
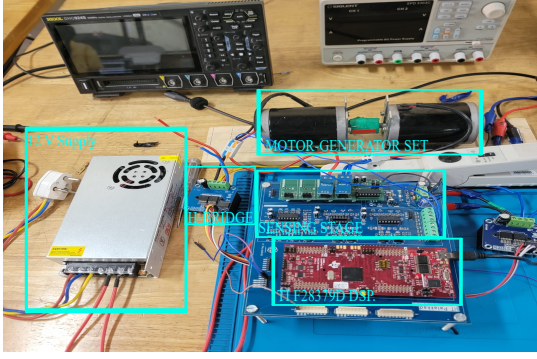

Virtual Model
V
Beta
wt
Wind Model
T_aero
T_TB,L
T_aero_pu
wt_pu
Rotor
Ht
Dt
Low speed shaft
Kls
Dls
wgb_pu
Gearbox
Hgb
Dgb
High speed shaft
Khs
Dhs
Compensation section
Ths_pu
wgb_pu
Tgen_pu
Dm
Hm
Dr
Hr
Tc
wB,L
V
Beta
wt
Tgen
Current controller
Tc
/Kt
Ia*
PI
d
Motor Driver
12V
Va
Motor-gen set
PMDC Motor
wm
Ia
Kb
Eg
Ra
La
Generator
Igen
Vout
A
Ro
Load
Motor speed estimator
1/Kb
Load Torque calculation
Kt
Sensing Stage
Ia
Igen
Vout


12V Supply
MOTOR-GENERATOR SET
H BRIDGE
SENSING STAGE
TI F28379D DSP